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Data Science

LendLoop – Campus Resource Sharing Application

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Guide
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SDG 12 – Responsible Consumption and Production

1. LendLoop promotes reuse of items instead of buying new one
2. Reduces waste and unnecessary consumption
3. Encourages sustainable usage of resources within campus
4. Supports efficient utilization of existing materials

1. Introduction

- **Problem Statement**

- Students often require resources such as books, calculators, lab equipment, etc.
- Purchasing items for short-term use can be expensive.
- There is no organized system within the campus for sharing resources.

- **Solution Proposed**

- Develop LendLoop, a mobile application for campus resource sharing. Students can lend, exchange, or request items from other students.
- The platform promotes efficient resource usage and collaboration.

2. Objectives

- To Develop a Centralized Resource Sharing Platform
- To Develop Efficient Borrowing and Lending Process
- To Develop Secure User Authentication
- Real-Time Data Management

3. Scope

- Can be implemented within college campuses
- Useful for students who want to borrow or lend resources
- Supports sharing of books, gadgets, and study materials
- Can be expanded to multiple colleges in the future.

4.Proposed System

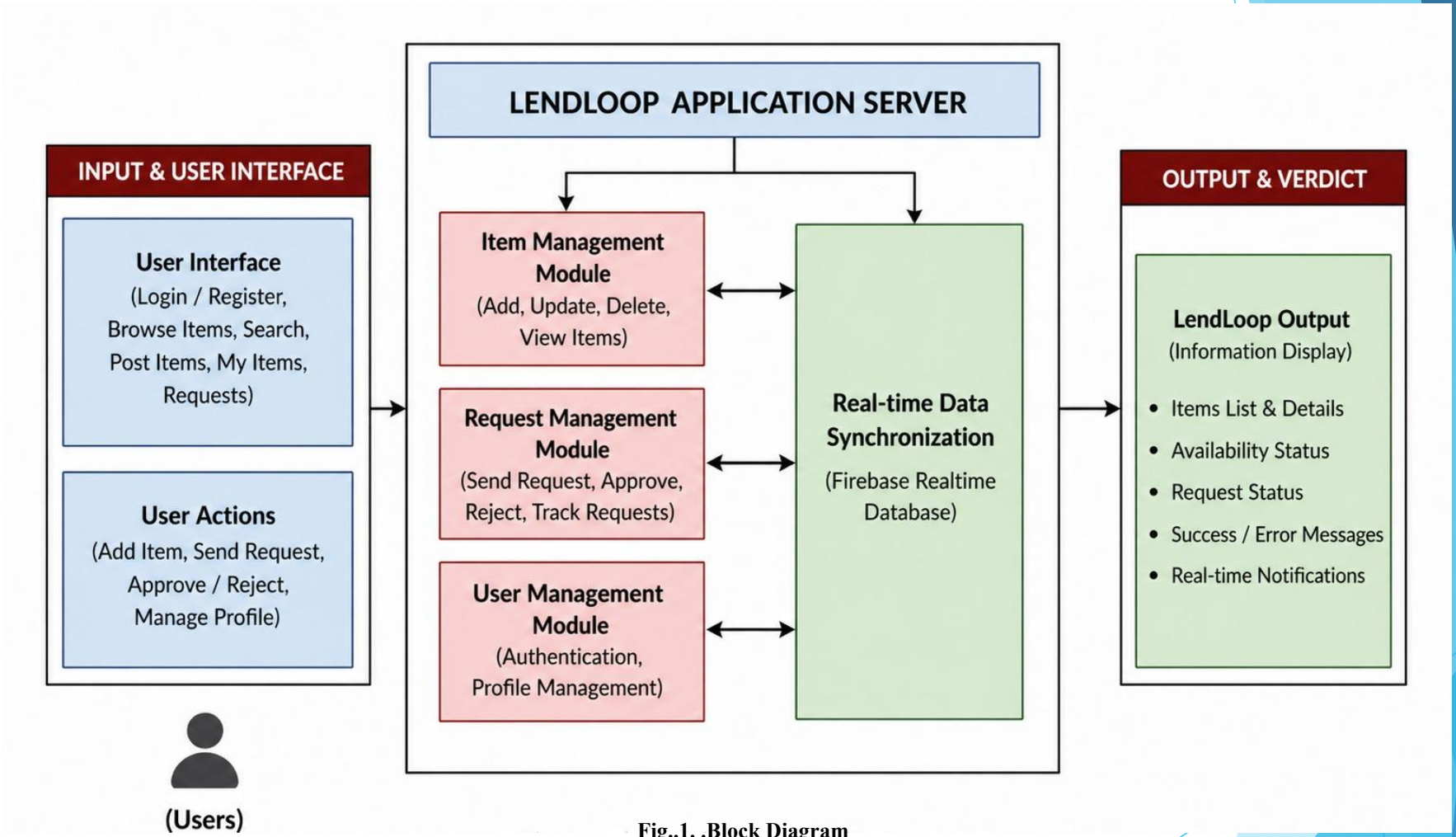


Fig..1. .Block Diagram

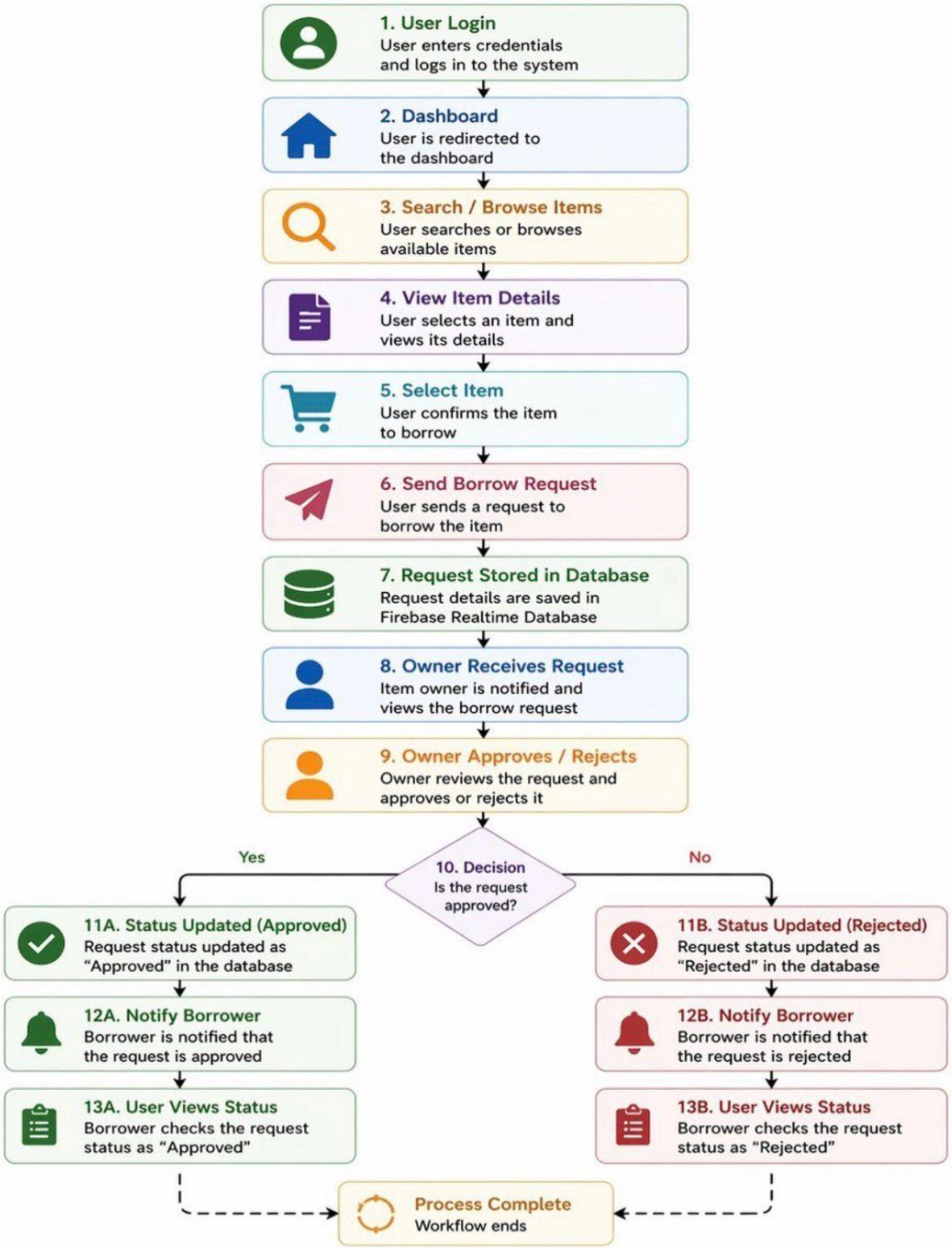


Fig..2. Workflow for lending an Item.

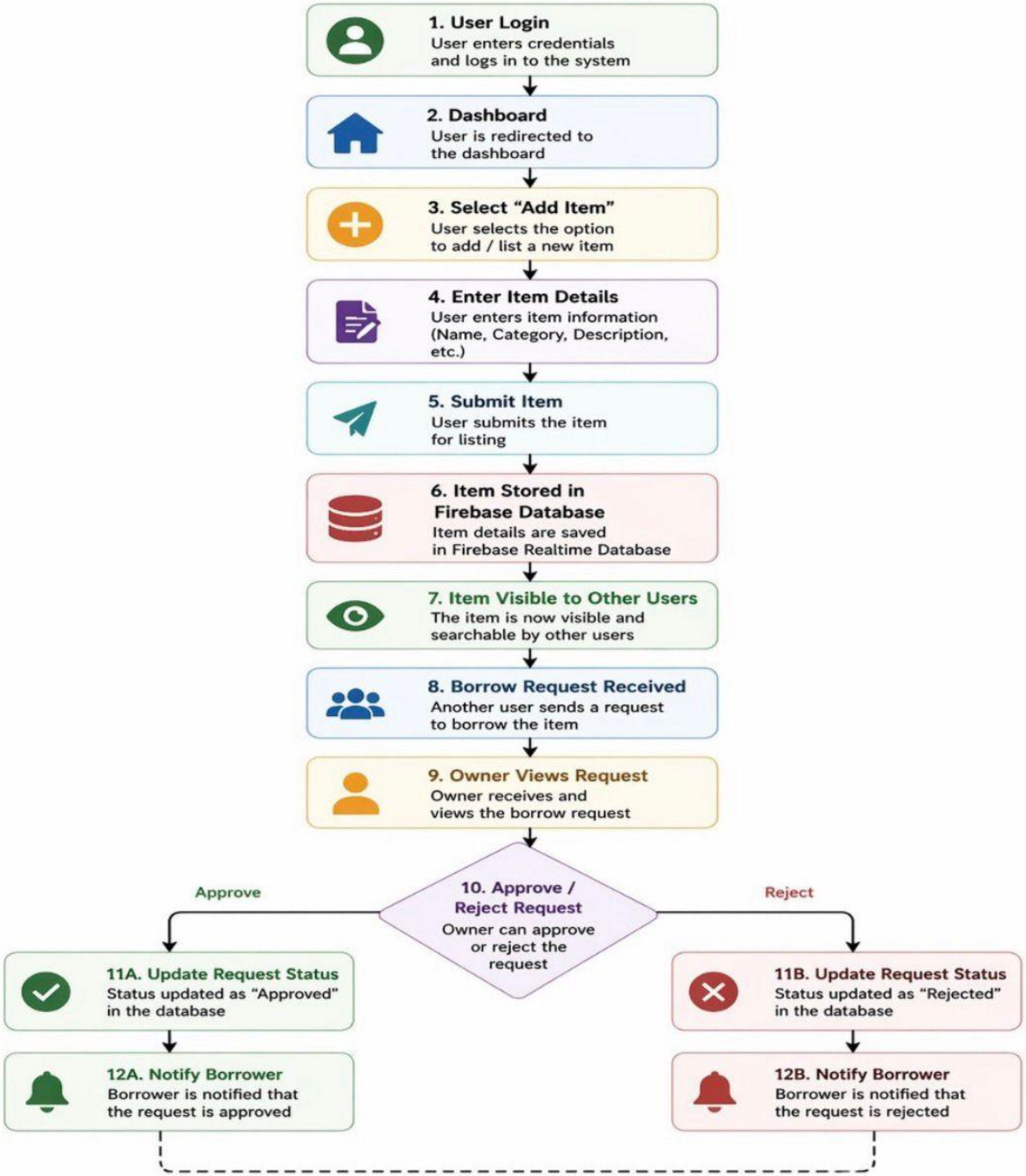
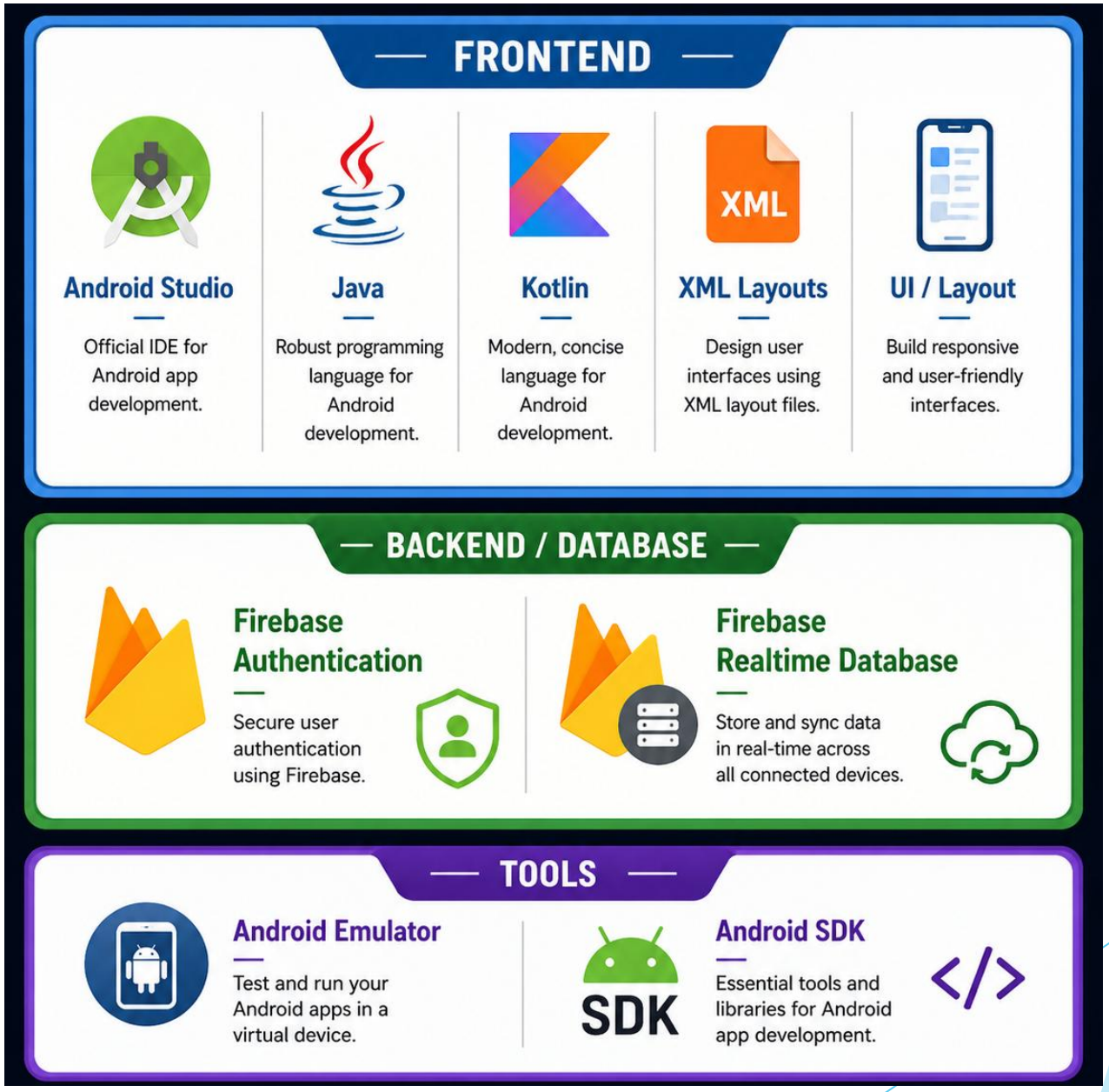


Fig.3. Workflow for searching and borrowing an item.

5. Technology Stack



6. Feature /Functionality

- **User Login System.**
  Users can log into the application using Firebase Authentication.

- **Dashboard Interface.**
  After login, users access a dashboard to navigate through the application.

- **Item Exchange Feature.**
  Students can exchange or lend items to other students.

- **Item Search Feature.**
  Users can search available resources within the application.

7. Outcome of Project

- Users can log into the application using Firebase authentication.
- Users can access the dashboard interface after login.
- Users can search available items in the system. Users can exchange items with other students.
- The application demonstrates a working prototype of campus resource sharing.

8.Results

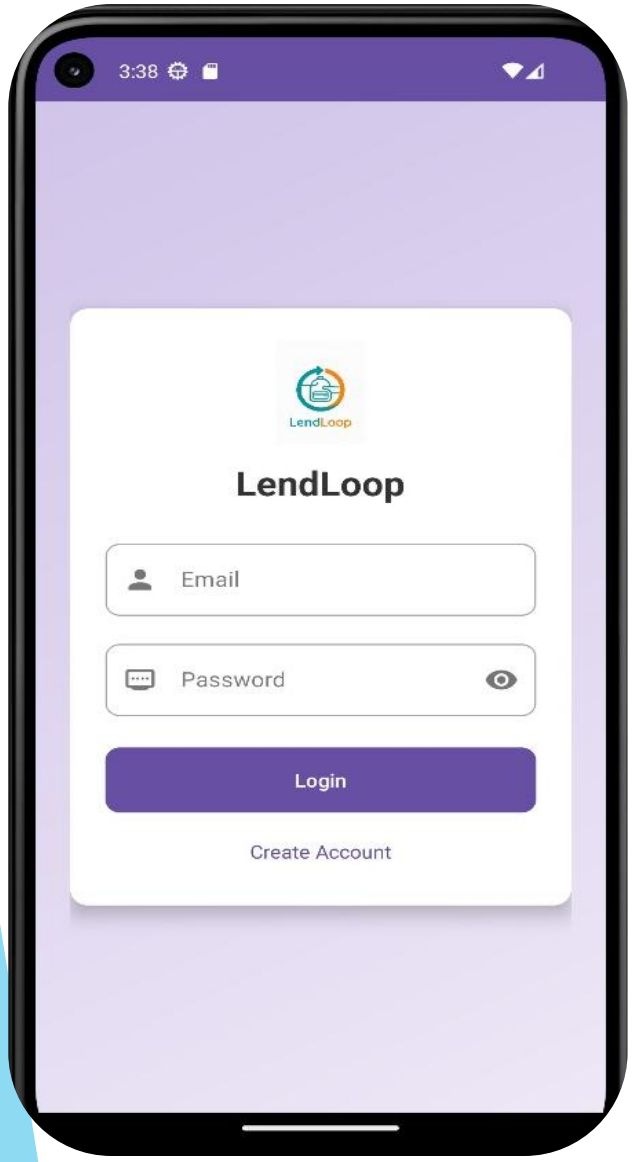


Fig.1.Login Page

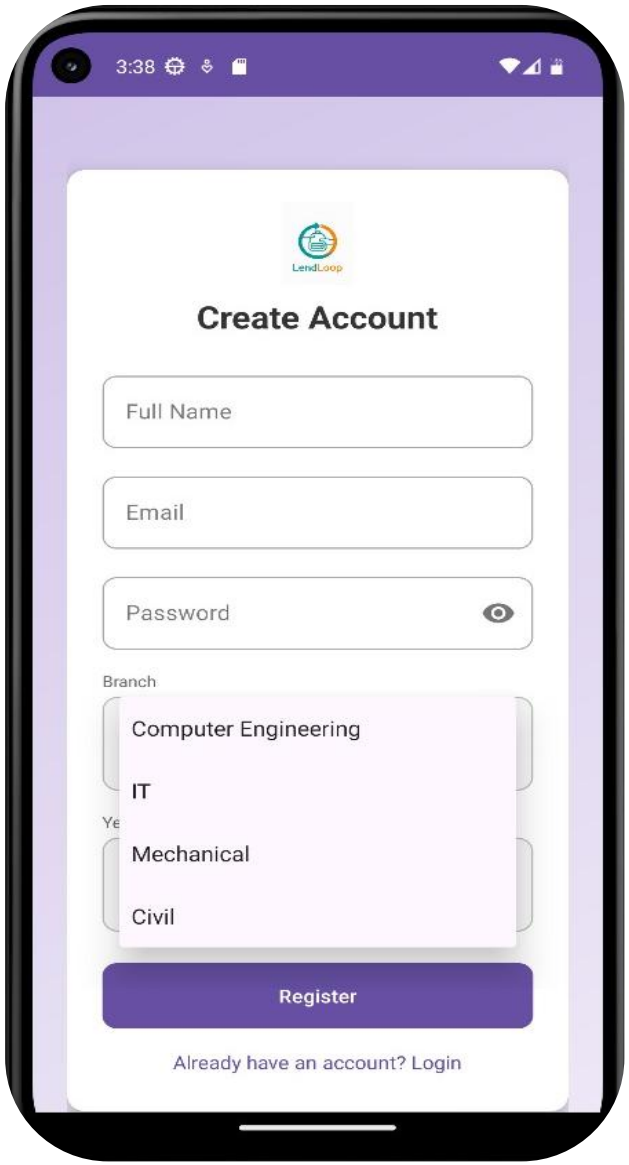


Fig.2.Register page

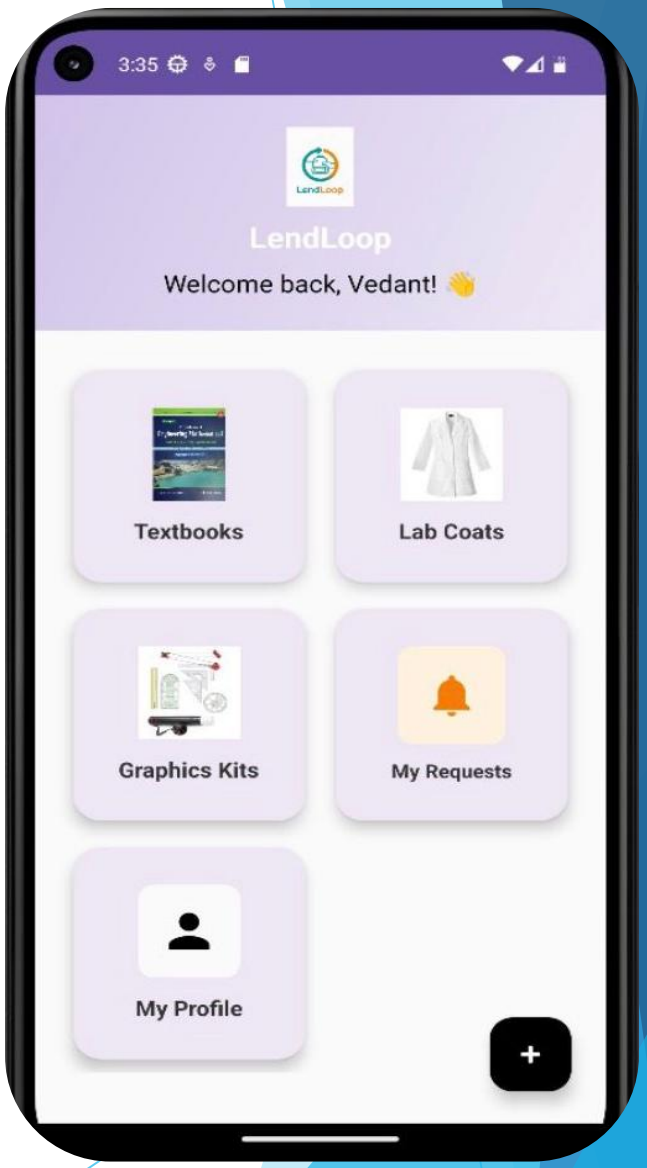


Fig.3.Dashboard

8.Results

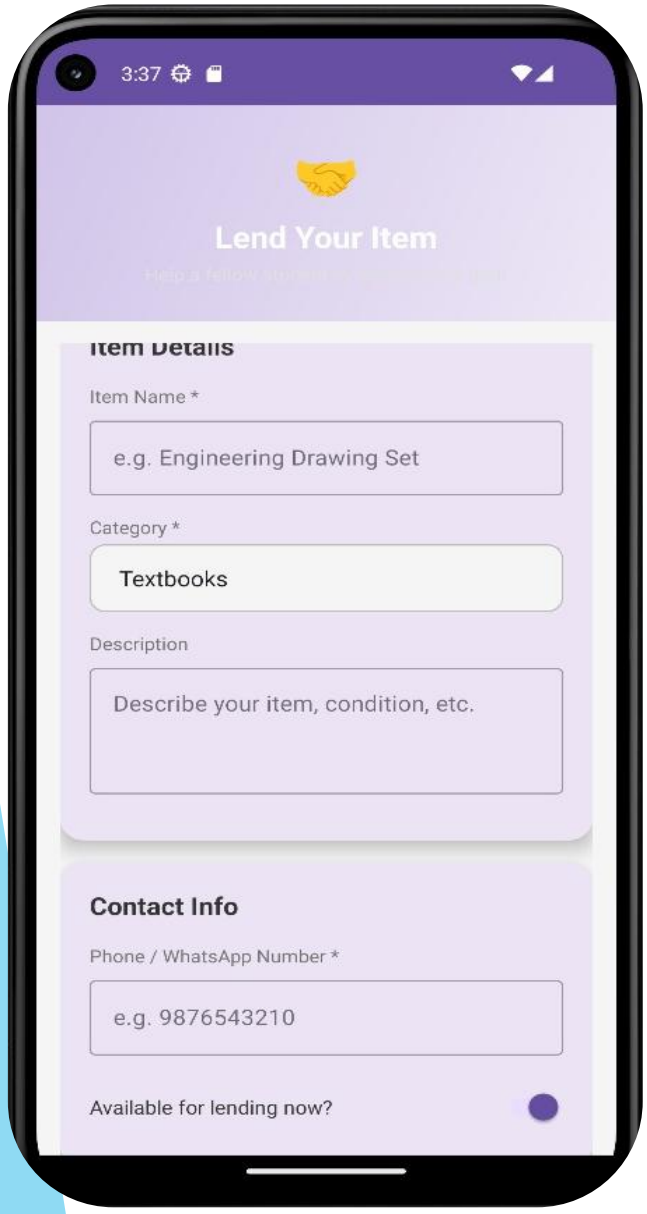


Fig.4.List Item for Lending

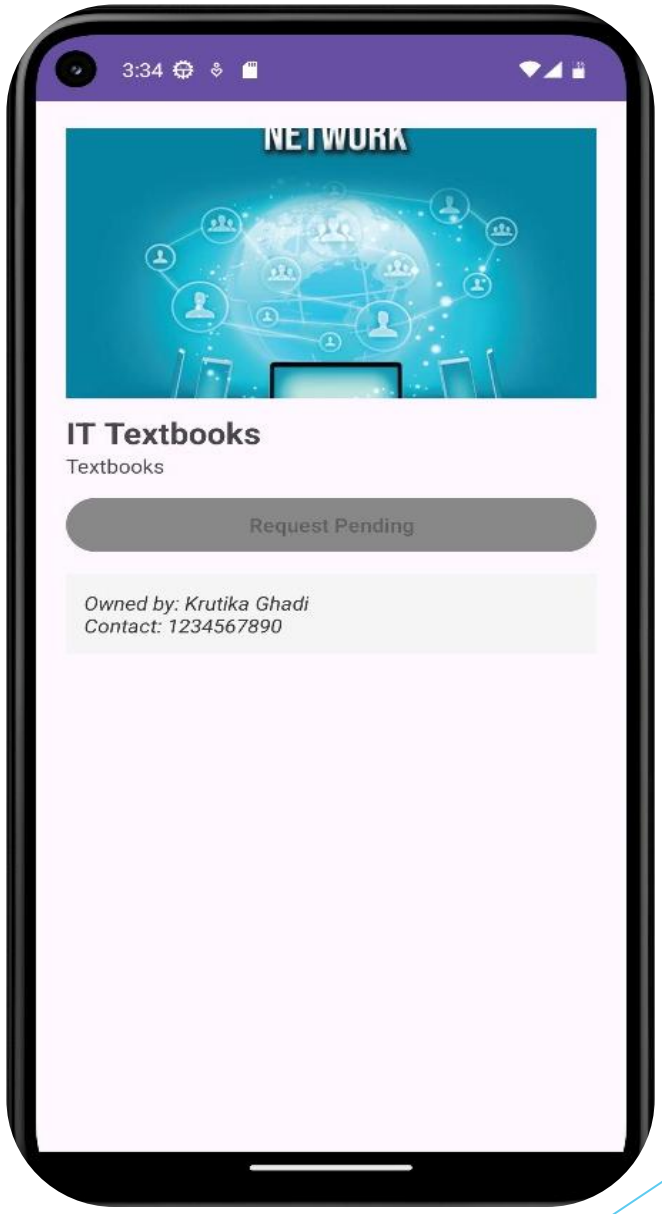


Fig.5.Selected Item

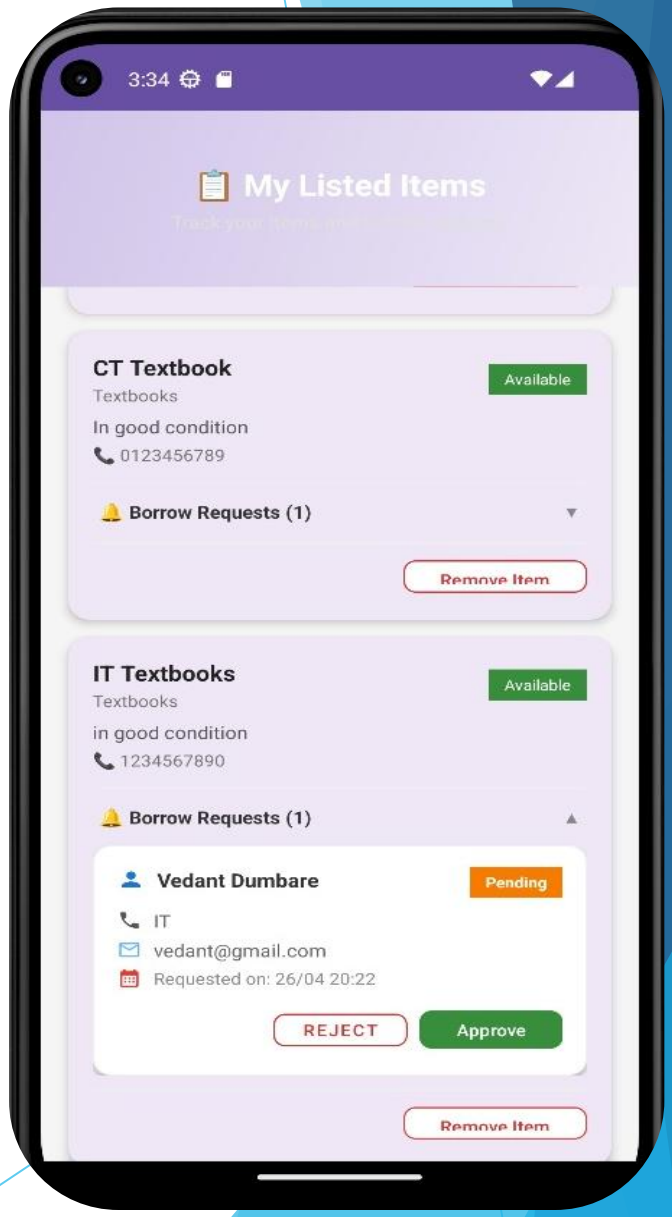


Fig.6. Listed Item to view request

Comparison with Existing Systems

Criteria	Existing Systems (OLX / Rental Apps)	LendLoop (Campus-Based System)
Target Audience	Open to general public	Restricted to verified college students
Trust & Security	Low (unknown users)	High (campus-based verified users)
Primary Purpose	Buying and selling items	Borrowing and lending resources
Cost Factor	Paid transactions	Free or minimal cost
Availability	Location-dependent	Easily available within campus
Ease of Use	Time-consuming search	Quick and simple access
User Verification	Limited or no verification	Authentication using college credentials
Focus Area	Profit-oriented marketplace	Resource sharing & sustainability

9.Conclusion

1. **Centralized Resource Sharing:** LendLoop provides a structured platform for easy borrowing and lending within the campus.

2. **User-Friendly & Efficient:** Allows users to list items, search resources, and manage requests with better visibility and control.

3. **Real-Time & Secure System:** Uses Firebase for instant data updates, secure authentication, and reliable performance.

4. **Impact & Future Scope:** Reduces costs, improves collaboration, and can be enhanced with features like payments, notifications, and multi-campus support.

10.References

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Thank You…!